

# SYMBIOSES OF GRASSES WITH SEEDBORNE FUNGAL ENDOPHYTES

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■ **Abstract** Grasses (family Poaceae) and fungi of the family Clavicipitaceae have a long history of symbiosis ranging in a continuum from mutualisms to antagonisms. This continuum is particularly evident among symbioses involving the fungal genus *Epichloë* (asexual forms = *Neotyphodium* spp.). In the more mutualistic symbiots, the *epichloë* endophytes are vertically transmitted via host seeds, and in the more antagonistic symbiots they spread contagiously and suppress host seed set. The endophytes gain shelter, nutrition, and dissemination via host propagules, and can contribute an array of host fitness enhancements including protection against insect and vertebrate herbivores and root nematodes, enhancements of drought tolerance and nutrient status, and improved growth particularly of the root. In some systems, such as the tall fescue *N. coenophialum* symbioses, the plant may depend on the endophyte under many natural conditions. Recent advances in endophyte molecular biology promise to shed light on the mechanisms of the symbioses and host benefits.

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## INTRODUCTION

Symbiosis is the typical state for large organisms, either plant or animal. The most widely studied symbioses are those that cause disease in the host. Other, more benign, symbioses are less obvious, but mutualistic symbioses are of major ecological and evolutionary importance. For example, most plants form mycorrhizae, interactions of roots with various fungi that facilitate nutrient uptake from soil (71). In the aerial parts of plants, endophytic and epiphytic fungi can be abundant, though inconspicuous (27, 28). The major subfamily of temperate grasses, the Poöideae, have many species possessing fungal endophytes that systemically infect much or all of the plant, some of which are also seed transmissible (141). This review focuses on the grass symbioses with the most widespread and beneficial of these endophytes, the *Epichloë* and *Neotyphodium* species (phylum Ascomycota, order Hypocreales, family Clavicipitaceae) (5, 86).

Note that the name *Neotyphodium* refers to asexual derivatives of *Epichloë* spp. (66). Here, we introduce “epichloë” as a general reference to these endophytes and use the Latin genus names only when specifying asexual or sexual species.

By definition, a mutualism must involve exchange of beneficial currencies between the partners, and must also exact mutual costs, but benefits must outweigh costs for both host and symbiont. Often, however, the relative benefits and costs depend on environmental or ecological contexts. For example, mycorrhizal symbioses show the most mutual benefit under low phosphate conditions (71). In contrast, benefits of clavicipitaceous endophytes to grass hosts have most often been observed under conditions of herbivory or parasitism (40), or in some cases under drought stress (101). Yet other benefits are also documented, including improved acquisition of phosphate and utilization of nitrogen (95, 101). The grass-endophyte symbioses demonstrate that benefits can be multifarious, and by coevolution of the partners an ever-expanding array of currencies may evolve.

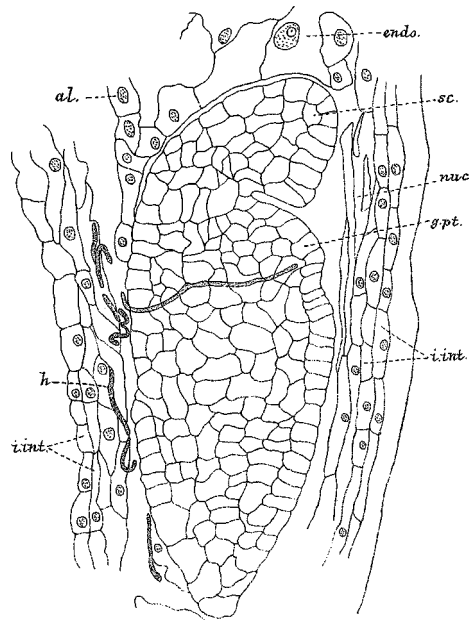
## COORDINATED LIFE HISTORIES

### Symbiotic Continuum and Life Cycles

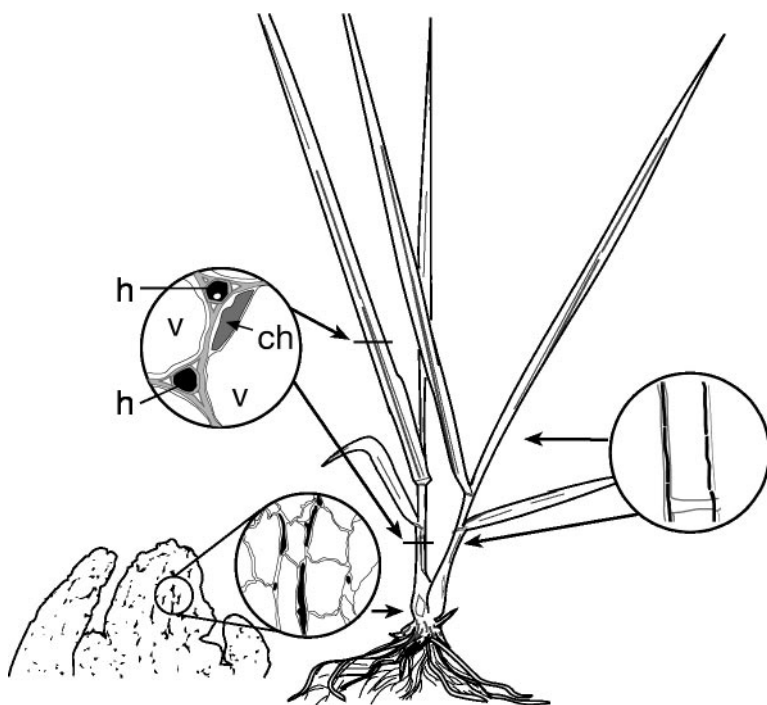
Symbioses of grass plants with epichloë fungi can be mutualistic, antagonistic, or exhibit both mutualistic and antagonistic characteristics. Where a symbiont sits on that continuum is strongly influenced by the symbiont life cycle, which in

turn strongly influences host reproduction. In the more antagonistic symbiots, host seed production is completely suppressed (choke disease) and the symbiont only transmits horizontally via sexually derived spores (ascospores) (150). In contrast, the more mutualistic associations are characterized by vertical transmission of the fungal hyphae via the seeds, which become the common diaspore for plant and symbiont. This represents an asexual form of endophyte reproduction. Completing the continuum are plant-epichloë symbiots that have both choked and symptomless inflorescences (130, 133), and exhibit both antagonistic and mutualistic effects.

Although *Epichloë* spp. had been known of for centuries, their strictly seed-borne relatives (in their inimitably subtle fashion) long eluded the attention of botanists. The first accounts of such endophytes were published near the end of the nineteenth century. Vogel (148) identified a densely interwoven layer of fungal hyphae in the remains of the nucellus between the aleurone and the seed coat of the grain in seeds of darnel (*Lolium perenne* subsp. *temulentum* = *L. temulentum*). In 1898 Guérin (67) suggested that the hyphal layer in the grain indicates mutualistic, rather than antagonistic, symbiosis. Among the studies of the darnel seed fungus that ensued, it was Freeman (62) who, in 1904, first detailed the entire coordinated plant-endophyte life history, observing that hyphae gain entrance to the embryo long before seed maturation (Figure 1). After seed germination the endophyte coordinates its growth with the above-ground plant tissues (Figures 2 and 3),



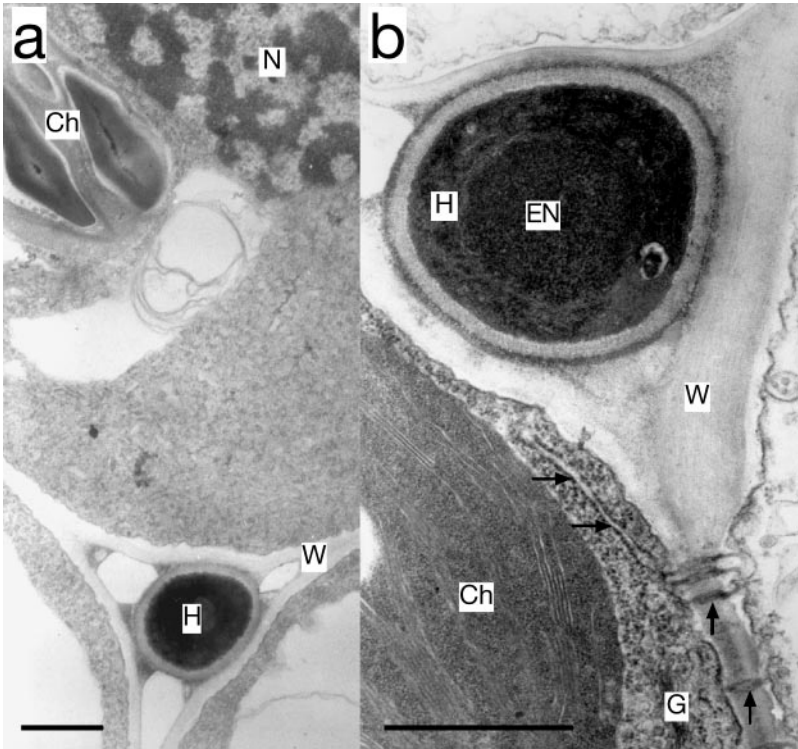
**Figure 1** Colonization by a *Neotyphodium occultans* hypha (h) of the embryo of *Lolium perenne* subsp. *temulentum* early in its development. Reprinted with permission (62).



**Figure 2** Endophyte growth in a grass plant. Bottom left: fungal growth in the true stem and leaf primordia. The fungal hyphae are shown darkly stained with osmium as they would appear in TEM (transmission electron microscopy). Upper left: A cross section of leaf or leaf sheath reveals hyphae (h) of the epichloë endophyte between host cells. Also shown are a chloroplast (ch) and vacuoles (v). Right: the endophyte as it appears in a leaf epidermal peel, stained for hyphae, which are arranged mainly along the longitudinal axis of plant cells. Unstained septa separate individual fungal cells, each of which bears a single haploid nucleus (not shown). Micrographs are at different scales. For comparison, epichloë hyphae are approximately 1–2  $\mu\text{m}$  thick.

colonizing lateral buds and, later, inflorescences. Vertical transmission of the asexual *Neotyphodium* spp. is remarkably efficient. In most natural associations the endophyte is present in essentially all seeds and, if the seed is freshly germinated, all progeny seedlings bear the symbiont (139). The hyphae never extend into roots or invade anthers. With the Freeman study (62) the systemic nature and maternal-line vertical transmission were clearly established for this representative of the clavicipitaceus endophytes.

In the 1930s Sampson (130, 131) made a key discovery concerning sexual endophytes. She compared an endophyte in *Festuca rubra* [now known as *Epichloë festucae* (92)] with *Epichloë typhina* on *Dactylis glomerata*. Whereas the latter



**Figure 3** Electron micrographs showing hyphae of (a) *Neotyphodium aotearoae* and (b) *N. australiense* in transverse leaf sheath sections of *Echinopogon ovatus*. Annotations are as follows: hypha (H), endophyte nucleus (EN), host nucleus (N), chloroplast (Ch), host cell wall (W), Golgi apparatus (G), plasmodesmata (vertical arrows), and endoplasmic reticulum (horizontal arrows). The plant cells appear normal, with no obvious damage or response to the endophytes. Scale bars represent 1  $\mu$ m. Reprinted with permission (108).

represents a textbook example of choke disease affecting all inflorescences, the *F. rubra* symbiont exhibits both the ability to cause choke and the ability to disseminate via seeds in a fashion similar to the strictly seedborne endophytes of perennial ryegrass (*L. perenne* subsp. *perenne*) and darnel. Remarkably, *E. festucae* can manifest its sexual cycle or asexual cycle on different tillers of the same plant, a characteristic later shown for most *Epichloë* spp. in poïd grasses (Figure 4). This finding connected the asexual endophytes with their sexual relatives.

The *Epichloë* sexual cycle (Figure 5) commences with accelerated fungal proliferation when the host forms inflorescence primordia (80). Grass internodes

elongate, raising the immature inflorescence, which is initially surrounded by a leaf sheath. Immediately prior to the stage when the inflorescence would emerge from the boot, the white endophyte mycelium proliferates specifically on the leaf sheath surrounding the young inflorescence, and mycelia thoroughly ramify the inflorescence. This fungal structure is the young stroma, which is loaded with conidia (mitotic spores). With stroma formation, the underlying inflorescence fails to emerge, and no seeds are produced on that tiller. If conidia (acting as spermatia) of one mating type are transferred to a stroma of opposite mating type, fruiting structures develop over the next three to six weeks. Eventually, fruiting bodies (perithecia) develop with scores of sacs (asci), each with eight ascospores. The ascospores are ejected into the air and mediate infection of neighboring plants (19) or developing seeds produced on neighboring plants (39).

### The Third Mutualist and Fungal Pollination

The *Epichloë* spp. require fly “pollinators” (Figure 5). The opposite mating types, mat1 and mat2, are cross-fertilized by flies of the genus *Botanophila* (order Diptera, family Anthomyiidae) (1, 24, 72, 82). Female flies transfer spermatia within their gut as they feed and oviposit. Immediately following oviposition, the fly fertilizes the stroma by a stereotypical fly behavior: She drags her abdomen while excreting feces on the stroma surface (24). Her larvae grow by feeding on the developing perithecial stroma and ultimately drop to pupate in the soil. This appears to be an obligate mutualism having much in common with associations between plants and their pollinating parasites (2, 21, 116, 146).

At least five described species are associated with *Epichloë* hosts in Europe (49, 72, 105). Additional fly taxa genotypically distinct from the European species were identified in North America (A. Leuchtmann, unpublished data). Two or more different *Botanophila* species can co-occur at a particular locality (49).

This unique association of *Epichloë* with *Botanophila* spp. is a delicately balanced symbiosis. The *Epichloë* fungi benefit from the flies as reliable vectors of spermatia, and the larvae depend on fertilized stromata as food (24). Larvae on unfertilized stromata seldom reach pupation (A. Leuchtmann, personal observation). As in pollinating parasites of flowering plants, the fly mutualism imposes a cost to the fungus by reducing the potential output of ascospores (149). However, flies do not overly exploit stromata, and the net effect of *Botanophila* on ascospore production is clearly positive (23, 24). One possible explanation for how the symbiotic balance is maintained is the observation that greater fly visitation with multiple eggs laid on a single stroma increases larval mortality. The causes for mortality are unknown but could be diseases or the parasitoids that commonly emerge from *Botanophila* pupae (82; A. Leuchtmann, personal observation).

The nutritional dependence of *Botanophila* larvae on fertilized stromata suggests that females should maximize cross-fertilization of stromata. This could be promoted through species-specific visitation habits by flies. Analysis of ascospore

progeny from experimental field plots indicates prevalence of specific matings between stromata of the same host *Epichloë* species, suggesting that fly visitations can be species-specific (89, 90). Genotypes of spermatia contained in feces of individual flies support this hypothesis (22). Although most flies carry a mixture of spermatia of different *Epichloë* spp., individual flies tend to carry spermatia predominantly from one species. Thus, flies may adopt “majoring” and “minoring” specificities for their host fungi. Fly selectivity may be one of the mechanisms promoting reproductive isolation among some *Epichloë* species or host races.

## REGULATING ENDOPHYTE GROWTH IN PLANTA

A crucial characteristic predisposing epichloë endophytes to evolution of mutualism is their strictly controlled intercellular growth throughout the host plant. The hyphae never breach host cell walls nor develop conspicuous feeding structures such as haustoria or arbuscules (62, 75, 118). Some but not all epichloë endophytes colonize vascular bundle tissues without ill effects (33). Only in association with stroma development are host cells typically damaged. White (151) suggests that the collapse of epidermal cells in the stromal leaf helps direct transpiration and, with it, nutrient flow through the stromal tiller.

Within the grass leaf the endophyte follows very distinct basal to apical hyphal concentration gradients (35, 73, 77). The hyphae appear to originate from profusely branching mycelium in the basal meristems (34), required to form new leaves, tillers, and inflorescences (34, 62, 118, 144, 151, 152). In the expanding and mature leaf tissues the hyphae are rarely branched and are arranged mainly along the longitudinal axis of adjacent plant cells. What causes the transition from highly branched hyphae in the plant base to single, thread-like hyphae in the leaf sheaths and blades is unknown, but may involve physical restrictions to branching as well as biochemical cues released from the surrounding plant cells. Within the plant, the meristems represent strong sinks providing a nutrient-rich environment for hyphal growth. In such an environment the endophyte may not need to alter or damage the host cells to obtain the necessary nutrients for profuse growth. Several enzymes putatively involved in nutrient acquisition were recently discovered (84, 85, 93, 94, 111, 123), though some of these could also be involved in hyphal growth and branching, and perhaps even in suppressing potential host defenses.

It has been suggested that colonization of newly developing leaves occurs by apical growth of hyphae into the elongating leaf (136). Christensen and colleagues (74) questioned this model, noting that hyphae appear to be intimately attached to host cell walls and the intercellular matrix, which are often interdigitated with the outer cell wall of the hyphae (see Figure 3) (118). Also, hyphae in the tips of expanding leaves appear to be older, not younger, than those at the basal parts of the same leaf. Christensen hypothesized that the hyphae undergo intercalary elongation as leaf cells also expand, thus preventing destruction of the hyphae (74). This proposal is intriguing because intercalary growth is very rare in fungi.

Endophyte growth is strictly synchronized with grass plant growth; hyphae actively grow in expanding leaves but cease to grow as the leaf completes elongation (144). Such closely coordinated growth also characterizes embryo colonization (62). The exception is the extensive proliferation and ramification during stroma formation to initiate the fungal sexual cycle. Even this stage appears to be highly regulated because it occurs only during flowering and only between the flag leaf collar and its underlying node (80, 132, 133, 152).

Because epichloë colonize all above-ground organs of the plant, they must adjust their growth rates and forms as the plant tissues differentiate. The most dramatic example of such a developmental switch is stroma initiation versus benign infection of developing inflorescences leading to seed transmission. Colonization of the plant and endophyte differentiation is most likely governed by multiple factors: biochemical changes such as phytohormone or metabolite concentrations, and/or physical changes such as movement of dividing or elongating plant cells (144). The observed host specificity of grass endophytes (37, 81) strongly suggests that host and symbiont communicate by specific signals.

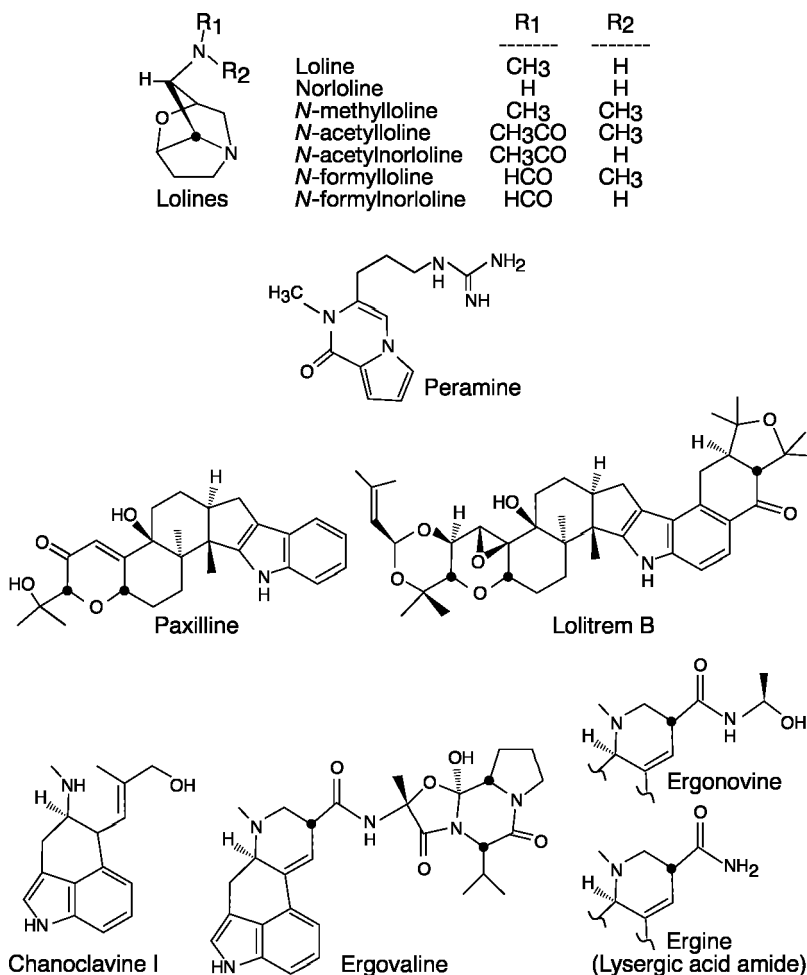
## EFFECTS ON OTHER ORGANISMS

### Anti-Insect Activities

Endophyte symbiosis exerts effects on numerous species including herbivorous and parasitoid insects, nematodes, granivorous and herbivorous birds and mammals, mycorrhizal and pathogenic fungi, and neighboring plants. A remarkable characteristic of many epichloë is their ability to produce several distinct classes of biologically active alkaloids (Figure 6). Activities of epichloë alkaloids on insects are well documented, and have been the subject of several major reviews (26, 41, 43, 44, 138). The neurotropic activities of lolines, and the activity of peramine as a feeding deterrent, can significantly enhance competitiveness of grasses with endophytes that produce these alkaloids. One of the most dramatic examples involves the Argentine stem weevil (*Listronotus bonariensis*), a devastating pest of perennial ryegrass in New Zealand. If undeterred, this exotic pest feeds on the crowns, and in much of New Zealand can eradicate entire stands. *Neotyphodium lolii* effectively protects the grass from this and several other insect herbivores (120), and the endophyte alkaloid, peramine, is primarily responsible for feeding deterrence (128). Peramine production in various endophyte-infected grass species also correlates with activity against the aphid *Schizapus graminis* (140).

Loline alkaloids exhibit a broader range and more overt toxicity to insects than does peramine (25, 126). For example, lolines but not peramine in symbiota correlates with, and is genetically linked to, activity against the aphids *Rhopalosiphum padi* and *S. graminis* (140, 154), whereas no activity of peramine against *R. padi* was evident in no-choice feeding tests (140). Deterrent activities of lolines and peramine against sucking insects may also help to reduce infections by plant viruses vectored by those insects (45, 97). Loline alkaloid levels can increase





**Figure 6** Representatives of the four classes of alkaloids produced by epichloë endophytes of grasses, namely, 1-aminopyrrolizidine alkaloids (lolines), peramine (the only known pyrrolopyrazine), indole diterpenes (here represented by paxilline and lolitrem B), and ergot alkaloids (ergovaline, ergonovine, and ergine). Also shown is chanoclavine I, a clavine alkaloid precursor to the ergot alkaloids.

dramatically in response to clipping (mock herbivory) (26, 50). Thus, lolines represent an inducible defense in the symbiotum.

## Effects on Vertebrates

Two other classes of endophyte alkaloids, indole diterpene and ergot alkaloids, possess both anti-insect and antivertebrate activities, and are implicated in

livestock problems such as ryegrass staggers and fescue toxicosis (83). In the 1930s, tall fescue (*Lolium arundinaceum* = *Festuca arundinacea*) was bred and widely disseminated in the United States, touted as excellent forage with considerable longevity, stress tolerance, and capacity to prevent soil erosion. However, by the mid-1970s the problem of fescue toxicosis in cattle and other livestock had been recognized, with symptoms resembling ergot poisoning caused by *Claviceps purpurea*. Initially, researchers were unaware of the early work by Neill (112, 113) on endophytes in both tall fescue and perennial ryegrass. Bacon et al. (66) rediscovered the tall fescue endophyte now known as *Neotyphodium coenophialum*, and associated it with ergot alkaloids (12). Of particular note was ergovaline (96), an ergopeptine similar but not identical to toxins in *C. purpurea*-contaminated grain. An abundance of experimental evidence indicates that *N. coenophialum* produces alkaloids that cause the symptoms associated with livestock grazed on tall fescue (145).

In New Zealand and Australia, where perennial ryegrass is a major component of the agronomic system, toxicosis to grazing livestock was a long and persistent problem. There, the common symptomology mimicked paspalum staggers, caused by ingesting indolediterpene alkaloids produced by *Claviceps paspali* (47). On the heels of the tall fescue endophyte discovery, *N. lolii* was identified as the ryegrass endophyte causing staggers (60). The *N. lolii* strains most common in Australia and New Zealand produce indolediterpene alkaloids called lolitrems, referring to their tremor-inducing neurotropic activity (64, 127).

In addition to the problems associated with perennial ryegrass and tall fescue, endophyte-infected grasses native to several continents cause related symptoms. Most notable are drunken horse grass (*Achnatherum inebrians*) in Asia, sleepygrass (*Achnatherum robustum*) in North America, dronkgras (*Melica decumbens*) in South Africa, and *Poa huecu* (causing huecú toxicosis) in Argentina. The former two are associated with ergot alkaloids—specifically lysergic acid amide (ergine) and ergonovine—which induce stupor and aversion to future grazing (106, 117). The latter two examples are associated with tremors that suggest indolediterpene effects (46, 108, 119). Unlike ryegrass staggers, huecú toxicosis can be lethal (119).

Correlations of endophyte infection frequencies with grazing pressure on *Festuca* species were investigated in two distinctly different ecosystems. In the southwestern United States, most comparisons indicated no significant differences in endophyte incidence between paddocks from which large grazers had been excluded for 5–14 years versus adjacent stands that remained available to wildlife (129). However, infection frequencies in almost all cases were extremely high, suggesting a different basis for endophyte maintenance such as drought tolerance effects (110).

For *F. rubra* on the Scottish islands of St. Kilda and Benbecula, infection frequencies correlate with grazing from feral Soay sheep, supporting the hypothesis that the endophyte enhances survivability under grazing pressure, perhaps by deterrence (14). Also, population crashes in the feral sheep may be due in part to epichloë toxicity (14). In the same study, ergot alkaloid levels greatly increased in *F. rubra*-endophyte symbiota upon mock herbivory (clipping), again showing that an epichloë metabolite can represent an inducible plant defense.

Endophyte effects on populations of small mammals are well documented, as are deterrent effects on granivorous birds (reviewed in 43). Clay & Holah (42) observed that tall fescue with *N. coenophialum* dominated the field populations of plants to a much greater extent than endophyte-free tall fescue, and attributed this to reduced herbivory on the *N. coenophialum*-infected grass. Supporting this hypothesis, vole reproduction was significantly reduced in plots with endophyte-infected versus endophyte-free tall fescue (61). As *L. arundinaceum*, mainly *N. coenophialum*-infected, has become dominant in large areas of the continental United States, populations of small mammals appear to be negatively affected (48, 65).

## Effects on Nematodes

Among the most dramatic effects of *N. coenophialum* in tall fescue is reduced parasitism by root-knot nematodes (*Meloidogyne marylandi*) and the migratory nematode *Pratylenchus scribneri* (58, 79). This is interesting because the endophyte is completely absent (or nearly so) in roots (10, 75), so whatever is responsible for antinematode activity must either be translocated from the endophyte-infected tissues or induced in the plant by endophyte. In fact, an induced structural change in the root—namely, thickening of endodermal cell walls—might reduce the ability of *M. marylandi* to penetrate the steele, induce giant cells (which provide nutrients to *M. marylandi*), and reproduce (68, 79). The reduced parasitism helps enhance effects associated with drought tolerance, namely, osmotic adjustment in growing points of the plant, and root growth (58).

Unlike the root-knot nematode, *Pratylenchus scribneri* penetrates roots of *N. coenophialum*-infected and endophyte-free tall fescue equally well, yet *P. scribneri* reproduction is also inhibited by endophyte presence (79). Thus, antinematode activity may be due to both a physical barrier and other mechanisms not yet elucidated.

## Effects on Fungi and Plants

Reports of *in symbio* antifungal effects of endophytes are rare. Some protection of seedlings against the soilborne pathogen *Rhizoctonia zeae* has been documented (69), and *E. typhina* protects against *Cladosporium phlei* (137). Also, there are inconsistent indications that tall fescue allelopathy against clover may be enhanced by *N. coenophialum* (143). These aspects need greater attention in future.

## MOLECULAR BIOLOGY OF ENDOPHYTE SECONDARY METABOLISM

In some symbiota, lolines accumulate to very high levels, up to 2% plant dry mass (25, 26), probably well exceeding endophyte biomass. Recently, *N. uncinatum* cultures were demonstrated to produce comparable levels of lolines (17a).

Additionally, Mendelian and molecular analysis identified polymorphism at a single locus (*LOL*) in *E. festucae* that determines whether or not lolines are produced (154). A *LOL* locus gene was found to be related to genes for synthesis of cystathionine (a methionine precursor), and to dihydrorhizobitoxine synthase (142a). Further analysis of *LOL* in *E. festucae* and *N. uncinatum* indicated a cluster of at least nine genes, all appearing to be unique to loline alkaloid producing strains, and most having significant similarity to metabolism genes (142a; M.J. Spiering, H.H. Wilkinson & C.L. Schardl, unpublished data). These gene relationships, along with results of precursor feeding studies (J.D. Blankenship & C.L. Schardl, unpublished data), suggest a pathway for loline alkaloid biosynthesis that differs substantially from any known metabolic pathway. Thus, lolines represent a fungal trait characterized by a specialized biosynthetic pathway culminating in an abundant and potent defense of the host plant.

An etiological role for ergot alkaloids in fescue toxicosis seems likely but has yet to be proven. Recent advances in molecular genetics of ergot alkaloid biosynthesis open the door to more rigorous tests. The *dmaW* gene for the first step in clavine and ergot alkaloid biosynthesis has been cloned, first from *Claviceps fusiformis* (147a), then from a *N. lolii* × *E. typhina* hybrid endophyte of *L. perenne* (148a). The *lpsA* gene for the penultimate step in ergovaline production was cloned from the same endophyte (115a). Directed mutagenesis of these genes in the endophyte has confirmed their roles in ergot alkaloid biosynthesis (115a, 148a). Symbiota with such mutant and corresponding wild-type endophytes can be used to investigate the roles of these alkaloids in livestock toxicoses and other host fitness enhancements.

## EFFECTS ON HOST PHYSIOLOGY

### Effects on Growth and Abiotic Stress Tolerance

Many observations, made in controlled environments on single cultivars and natural ecotypes of tall fescue, meadow fescue (*Lolium pratense* = *Festuca pratensis*), and perennial ryegrass, suggest that their epichloë endophytes (*Neotyphodium coenophialum*, *N. uncinatum* and *N. lolii*, respectively) have positive effects on plant growth. Enhanced biomass production, tiller numbers, seed production, and root growth have been reported (16, 54, 63, 87).

Substantial (>50%) stand losses in tall fescue were reported after removing the endophyte from this grass (122). These losses are typically associated with drought periods, and endophyte-infected tall fescue exhibits improved recovery after drought compared to endophyte-free tall fescue (7). It was proposed that grass endophytes, particularly *N. coenophialum* in tall fescue, affect plant water relations, nutrient acquisition, as well as allocation and photosynthetic assimilation (11, 63). Most investigations of epichloë effects on stress tolerance focus on osmotic adjustment, water relations, and drought recovery (17, 31, 55–57, 99, 110,

121), accumulation of drought-protective osmolytes in the grass tissues (124), and photosynthetic rates under water or heat stress (4, 103, 125). Under water stress, the tall fescue endophyte is also associated with a significant increase in cell wall elasticity as measured by bulk modulus tissue elasticity, and by turgid weight to dry weight ratio (TW/DW) (153). Likewise, *N. uncinatum* increases TW/DW in water-stressed meadow fescue (98). Endophytes can induce in tall fescue and meadow fescue increased root growth and longer root hairs, and decreased root diameter (100, 102).

Taken together, there appears to be a trend toward improved physiological responses of endophyte-infected grasses to adverse environmental conditions. However, studies with several grass species confirm complex interactions between endophyte status, plant genotype, water and nutrient availability, and spatial competition (3, 13, 17, 20, 29–31, 59).

Little is known about the biochemical basis for endophyte-induced changes that cause growth effects and increase stress tolerance. Production of phytohormones by the endophytes—e.g., auxin [indoleacetic acid (IAA)]—may play a role in plant growth alterations (53, 155). However, levels of free IAA in the whole plant are unaffected by endophyte infection (53). Nonetheless, localized or transient changes might occur in the plant but could be difficult to detect.

Osmotic protection is more likely than stomatal conductance to be involved in drought protection in tall fescue (9, 56, 57, 125), but reduced stomatal conductance might be important to conserving water in *Festuca arizonica*-*Neotyphodium* sp. interactions (110). Some speculation regarding osmoprotectants centers around the fungal loline alkaloids, which are abundant in those symbiota for which the endophyte has a documented and consistent positive effect on drought tolerance (101). Lolines fit several prerequisite criteria, being nontoxic to plant cells, highly water soluble, and generally increasing in response to heat or drought. However, it is unclear if lolines reach sufficient levels to significantly affect osmotic balance. If these alkaloids are involved, they might protect macromolecules from denaturation and/or scavenge reactive oxygen species associated with drought stress, possibilities not yet tested. Other potential osmoregulators and protectants are soluble sugars and sugar alcohols, produced by the endophyte, plant, or both (124).

## Effects on Photosynthesis

Although it is useful to evaluate endophyte effects on photosynthesis, one should consider that photosynthesis is not always a reliable predictor of the productivity of grasses and crops (88, 114). In fact, limiting the rates of photosynthesis improves forage quality in terms of nitrogen content and digestible dry matter (78). It is often difficult to distinguish the effects of fungal symbionts from other factors, such as plant species, age, and environmental conditions; photosynthetic responses are not uniform among grass species, and epichloë interactions

with plant genotype and age may be significant (4, 9, 15, 18, 52, 103, 107, 110, 115, 121, 125, 142). Most studies have measured rates of net photosynthesis and/or photochemical efficiencies in grass leaves, but little or no information exists about endophyte-induced changes in amounts and/or activities of photosynthesis enzymes. Identifying enzymes secreted by *epichloë* in planta, namely, invertases, glucanases and proteinases (84, 111, 123) might provide some leads. Higher invertase and glucanase activities can increase levels of sugar monomers, causing decreased enzymatic activities in the Calvin cycle (70, 109) and, consequently, decreasing rates of photosynthesis. Surprisingly, however, amounts of ribulose 1,5-bisphosphate carboxylase/oxygenase (rubisco), a key enzyme and rate-limiting step in the Calvin cycle (6), were unaffected by endophyte status (M.J. Spiering & J. Schmid, unpublished data), despite significantly lower rates of net photosynthesis of endophyte-infected grasses (142).

Given the wide variation in physiological responses of grasses to endophyte infection, using endophyte-grass associations well characterized for growth and photosynthesis and grown under carefully standardized environmental conditions would be essential in future experiments seeking to unravel the physiological responses to putative *epichloë* signals as well as biochemical and biophysical effects of the endophytes.

## EVOLUTION OF GRASS-ENDOPHYTE SYSTEMS

### Host Specificity

Natural host *Neotyphodium* symbioses and (in the vegetative state) host *Epichloë* symbioses exhibit little or no obvious host cell response to endophyte presence (see Figure 3). Possible explanations are either that the endophyte simply fails to trigger a response, or that the host response is actively suppressed. Although there is no evidence yet to distinguish these possibilities, abundant evidence indicates that compatible interactions are highly specific. In artificial inoculation studies endophytes have been exchanged between related host species, namely *L. perenne*, *L. pratense*, and *L. arundinaceum* (32, 81). Fungal mycelium was introduced into the apical meristematic region of seedlings or mature tillers. Some of the resulting novel associations elicited incompatibility reactions such as premature death of hyphae, death of host cells in the stem apex, and stunting of surviving tillers. Often osmiophilic material was deposited in host cell walls adjacent to the hyphae. Host specificity of *E. typhina* strains is heritable and under multigenic control (38).

### Epichloë Speciation and Co-Phylogeny with Hosts

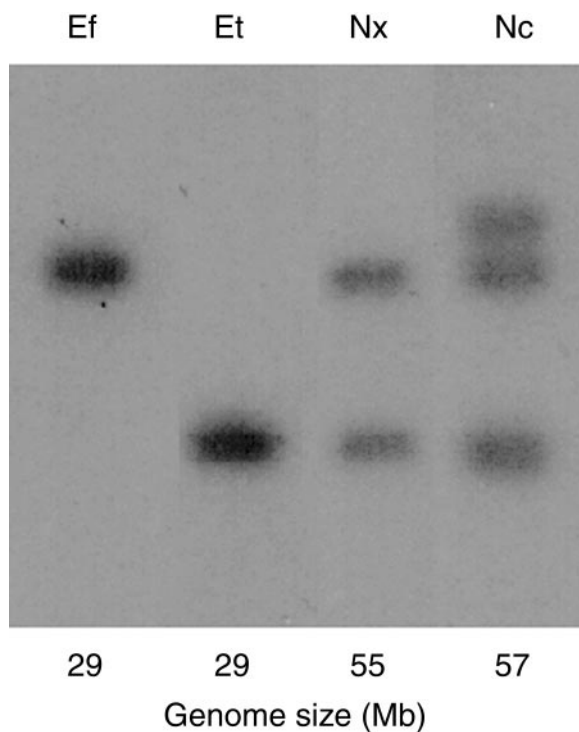
Current knowledge about the evolution of *Epichloë* and *Neotyphodium* species has been reviewed thoroughly in two recent publications (43, 135) and is summarized

here. Prior to 1993 the *Epichloë* spp. on grasses of subfamily Poöideae were all classified as *E. typhina*. Since then, nine new species have been described with the aid of molecular phylogenetic analysis and mating tests. All the new species exhibit limited host range, usually restricted to individual host genera or related genera within a host tribe. The exception is *E. typhina* as currently circumscribed, which includes interfertile strains associated with many grass species in at least three tribes. Most *Epichloë* species are capable of vertical transmission. Of these, the most intensely studied is *E. festucae*, for which vertical transmission dominates (8) and which has strong mutualistic character (133). Molecular phylogenetic analysis suggests that the seed-transmissible *Epichloë* spp. have a history of cocladogenesis (diffuse cospeciation) with the Poöideae (134). If so, then the grass-epichloë system is approximately 40 million years old.

### Interspecific Hybrids

Some asexual endophytes appear to have evolved from their *Epichloë* spp. ancestors by losing the capability to produce stromata and thereby initiate the sexual cycle. However, most asexual endophytes analyzed to date are interspecific hybrids, with combined genomes or partial genomes of two or sometimes three ancestors (135, 147). The genotypes of certain hybrids in relation to nonhybrid asexual endophytes suggest that hybridization followed loss of sexual expression (though in some cases the process may have caused the change to a strictly asexual state). The common tall fescue endophyte, *N. coenophialum*, is a dramatic example of an interspecific hybrid, having three *Epichloë* species in its ancestry (147). This is illustrated, for example, by its three  $\beta$ -tubulin alleles (Figure 7), which are in contrast to the single allele for this gene found in every known sexual species. One of the *N. coenophialum* alleles is derived from *E. festucae*, another from *E. typhina*, and another from a third species yet to be identified. Hybrid endophytes appear to retain much of their ancestral genomes, having substantially larger genome sizes than sexual *Epichloë* spp. (Figure 7) (82b).

The abundance of interspecific hybrids in surveys of endophyte species and the dominance of some hybrids in host populations strongly indicate that they are selectively favored (135). The basis for that selection is not obvious, but it seems likely that hybridization provides similar benefits to sexual reproduction. These may include the ability to purge or mask deleterious mutations, or the pyramiding of genes for host fitness enhancements. The fates of epichloë lineages that are only transmitted vertically are inextricably bound to the fates of their individual host plants and maternal-line descendents of those plants, so that host fitness should be more important in their evolution than it is for horizontally transmissible epichloë. Another possibly selective factor is that hybridization enhances the pace of evolution to keep up with host evolution (note that many grass hosts are also interspecific hybrids). Without a sexual cycle to adapt to host evolution, clonal



**Figure 7** Southern analysis indicating single  $\beta$ -tubulin gene alleles in *Epichloë festucae* (Ef) and *E. typhina* (Et), two alleles in an *Neotyphodium lolii*  $\times$  *E. typhina* hybrid (Nx) from perennial ryegrass, and three alleles in *N. coenophialum* (Nc), a complex hybrid endophyte from tall fescue. Equal amounts of nuclear DNA were loaded in each lane. Genome size estimates were from quantitative Southern analysis and electrophoretic karyotyping (82b).

endophytes might not maintain these associations for long unless they have this additional evolutionary mechanism.

Evolutionary diversification of endophytes adds to the evolution of hosts to enhance the genetic diversity of symbiotes. As surveys of endophytes continue, it is apparent that grass species often harbor two or more seedborne endophytes of distinct genotype and ancestry (36, 135). Combined plant and endophyte hybridization results in symbiotes loaded with genes evolved in various ancestors under various ecological conditions. The tall fescue-*N. coenophialum* symbiont represents one of the world's most complex, heritable genetic systems. The plant has three nuclear genomes (as an allohexaploid) (76), two organellar genomes (plastid and mitochondrion), and an endophyte with contributions from three different *Epichloë* species (147).



## CONCLUDING REMARKS

The highly efficient heritability of many epichloë endophytes, and the dominance of endophyte-containing plants in populations of tall fescue (36), meadow fescue (36, 50), and other grasses, present the possibility that many symbiotes have evolved as grass-endophyte units over thousands or even millions of generations. In the process, the symbiotes have colonized diverse environments. For example, tall fescue with *N. coenophialum* naturally ranges from northern Europe into Morocco (51, 104). The broad range of endophyte benefits might be due to such long-term coevolution. The origins of these mutualisms are undoubtedly rooted in significant contributions by the symbionts of one or more benefits, such as protection from herbivory. However, after a long coevolutionary history the contributions of host and symbiont to many adaptations—such as to drought, nematodes, or low phosphate availability—should be much more entangled than if the plant were rarely or transiently associated with these endophytes. Intricate signaling systems would evolve, evidenced, for example, by the dramatic induction of endophyte alkaloids upon clipping of host leaves (50). Effects of these foliar endophytes on root architecture, apparently adaptive for drought tolerance and nematode resistance (58), also support this coevolutionary scenario.

Recent years have seen major advances in molecular biology and genetics of the endophytes and their hosts, placing these systems prominently among the models of mutualism in plants. Crucial aspects of the genetic systems and genome compositions of sexual and asexual epichloë endophytes have been elucidated, and the application of Mendelian and molecular genetic techniques have identified genes for alkaloids that play major roles in the mutualisms (115a, 142a, 148a, 154). Technologies for genetic and molecular genetic manipulations of the host grasses have steadily advanced as well (82a, 144a), and new functional genomic studies of endophytes and their hosts have been undertaken (54a, 131a). These advances position us to elucidate the specific mechanisms of grass-endophyte interactions and their mutualistic effects.

We now understand these systems sufficiently to start addressing some broader questions as well. For example, (a) Precisely what are the economic and environmental implications of grass endophytes, including but not limited to those in agronomic grasses? (b) How can the endophyte benefits best be used for agricultural and amenity grasses, and how can livestock detriments be minimized? (c) What is the ecological significance of grass endophyte diversity, and is this something that needs special attention for conserving germplasm and habitat?

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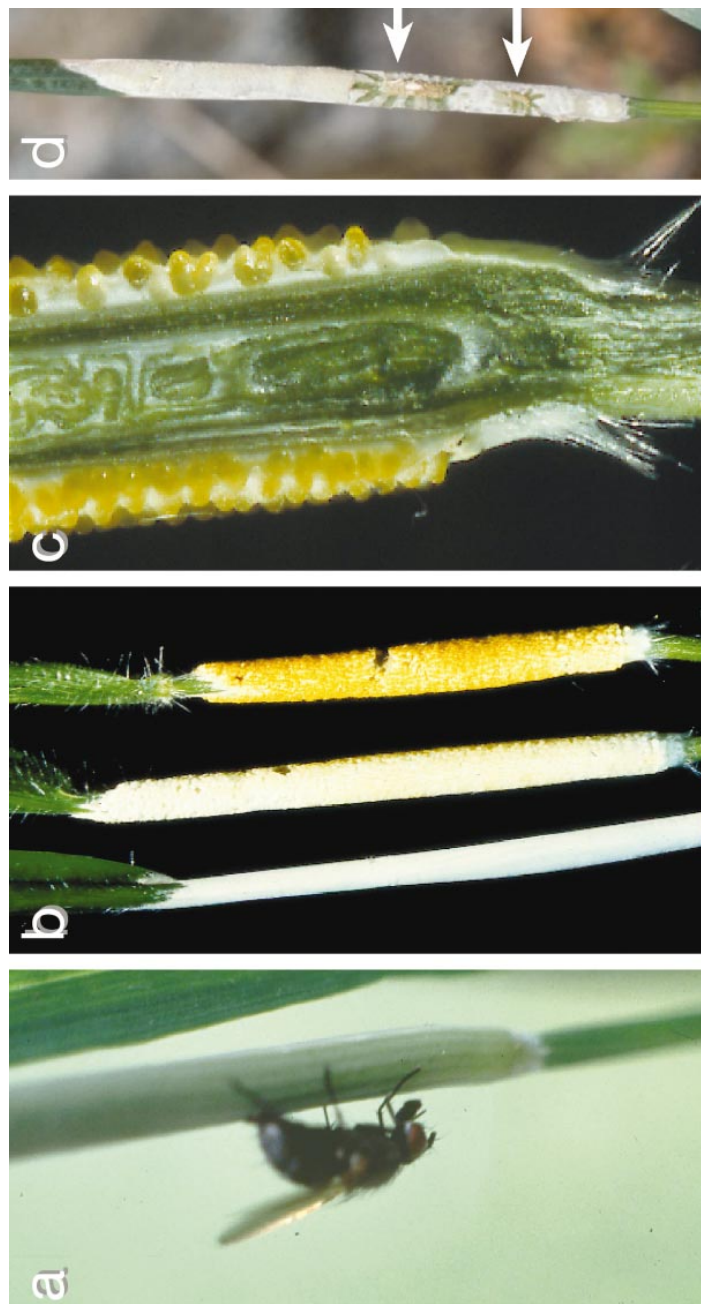


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**Figure 4** Simultaneous expression of diseased and benignly infected inflorescences on the same *Agrostis perennans* plant systemically infected with *Epichloë amarillans*. The normally developing inflorescences (*above*) produce seeds bearing the symbiont, whereas other inflorescences bear the sexual state of the fungus (*below*).



**Figure 5** Development of the *Epichloë* spp. sexual state, and the role of the *Botanophila* fly in cross-fertilizing the fungi. Panel a: stereo-typical behavior of the fly after ovipositing (see text). Panel b: three stromata in (from left to right) the early, middle, and late stages of maturation after fertilization by the fly. Note the distinct thickening of the stroma as perithecia (fruiting bodies) mature. Panel c: longitudinal section through a stroma. Embedded in the surface are yellow-orange perithecia. Fungal mycelium (white material) is apparent throughout the grass tissue under the stroma. Panel d: larval brood chambers (arrows) from which larvae emerge to feed on developing perithecia. Note the feeding tracks near the brood chambers.

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